DATA ANALYTICS PROJECT REPORT

Financial Performance Analytics Dashboard

Profitability, liquidity, cash flow and valuation analysis of Indian listed companies (FY2024)

Python | SQL Server | Power BI | DAX

3,887	₹128.13 L Cr	₹9.87 L Cr	7.7%
companies analysed	total sales	net profit	net margin

Figures: FY2024, non-financial listed companies, ₹ Crore unless stated (1 Lakh Crore = 100,000 Crore).

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1 Executive Summary

This project analyses the financial performance of **4,668 Indian listed companies** across four lenses: profitability, liquidity and solvency, cash flow, and valuation. The headline view covers the **3,887 non-financial companies** that reported FY2024 results. Raw data from 11 Kaggle CSV files is cleaned and reshaped in **Python**, modelled and queried in **SQL Server**, and presented in a five-page interactive **Power BI** dashboard with written insights and recommendations.

₹128.13 L Cr	₹9.87 L Cr	7.7%	3.1%	816
Total sales	Net profit	Net margin	Sales growth (YoY)	Loss-making companies

Key findings

- **Profits are concentrated.** The top 10 companies earn 39.5% of total net profit, and 259 large caps (market cap above ₹20K Cr) earn 87%.
- **Profitability rises with size.** Net margin is 8.8% for large caps and -0.6% for micro caps. 816 companies (21%) are loss-making.
- **Balance sheets are conservative at the median** (debt-to-equity 0.24, current ratio 1.81). By Altman Z-score, 78.1% of companies are Safe and 8.5% are in Distress, with risk concentrated in micro caps (70.9% Safe versus 91.5% for large caps).
- **Cash generation is strong.** Operating cash flow of ₹15.99 L Cr equals 162% of net profit, yet 70 profitable companies report negative operating cash flow.
- **Valuations rise with size.** Large caps trade at a median 55.4x P/E and 8.16x P/B versus 30.4x and 1.95x for micro caps.

> **What this project demonstrates:** an end-to-end analytics workflow, from messy wide-format source files to a validated SQL model and a decision-ready dashboard, with careful handling of outliers, unit consistency and data limitations.

Project at a glance

Item	Detail
Domain	Financial performance analytics, Indian equity market
Data source	Kaggle: pacificrm/financial-sheets (11 CSV files, 4,668 companies, 107 industries)
Period	FY2023 and FY2024 (latest annual results)
Tools	Python (pandas, NumPy), SQL Server (SSMS), Power BI (DAX), Git/GitHub
Deliverables	Cleaning script, SQL schema + analysis + views, 5-page dashboard (.pbix), this report

2 Project Overview

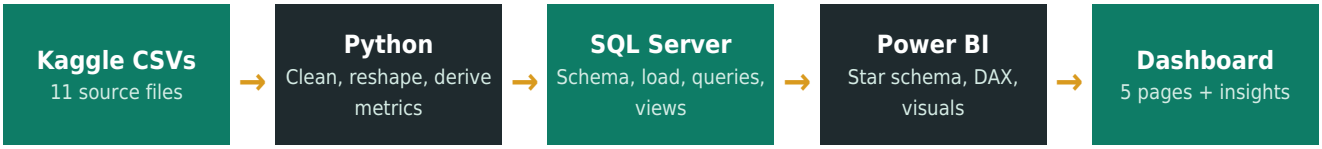
Objective

Build a professional, interactive dashboard that lets a reader judge the financial health and performance of Indian listed companies by industry and company size, and turn the numbers into clear findings and recommendations.

Business questions answered

- How profitable are Indian listed companies, and which industries lead?
- How concentrated are profits across company sizes?
- How strong are balance sheets: leverage, liquidity and interest coverage?
- Which companies show financial distress or high-leverage risk?
- Do reported profits convert into cash?
- How are companies valued relative to their size and returns?

End-to-end workflow



Tools and what each one did

Tool	Role in the project	Highlights
Python (pandas, NumPy)	Data cleaning and reshaping	Wide-to-long reshape, derived ratios, quality report, SQL-ready CSVs
SQL Server (SSMS)	Relational model and analysis	5 tables with primary and foreign keys, 9 analysis queries, 3 views, LAG / RANK / ROW_NUMBER
Power BI + DAX	Semantic model and dashboard	Star schema, 40+ measures, bookmarks, synced slicers, conditional formatting, custom theme
Git / GitHub	Version control and documentation	Scripts, .pbix, screenshots and this report in one repository

3 Dataset

The data is the Kaggle dataset **pacificrm/financial-sheets**: Indian listed companies on BSE and NSE. Every file has one row per company and a common, unique `join_key` (4,668 of 4,668 keys match across files). Eight of the 11 files were used.

File	Content used
Annual_P_L_1_final	Sales, operating profit, EBIT, interest, tax, net profit, EPS for the latest and preceding year; industry, market cap, price
Balance_Sheet_final	Debt, equity capital, reserves, total assets, current assets and liabilities, cash
cash_flow_statments_final	Operating, investing, financing and free cash flow for two years
ratios_1_final, ratios_2_final	ROE, ROA, ROIC, debt-to-equity, P/E, P/B, shareholding, 5-year average ROE
other_metrics_final	Current ratio, interest coverage, Altman Z-score, EV/EBITDA
price_final, t1_prices	1, 3 and 5-year price returns, T-1 price
Not used	Annual_P_L_2_final, Quarter_P_L_1_final, Quarter_P_L_2_final

What the data looks like

Characteristic	Value
Companies / industries	4,668 companies across 107 industries
Latest financial year	FY2024 for 4,507 companies (96.6%); the rest report an earlier year
Annual history	Two years only: FY2024 and FY2023 (no earlier annual P&L)
Structure	Wide format: each metric repeated as 'last year' and 'preceding year' columns
Financial-sector companies	628 flagged (banks, NBFCs, insurers)

Important discovery: the 'current' sales column equals the 'last year' column for about 97.6% of rows, so the file holds **two** annual periods, not three. This was confirmed against known figures (for example Infosys FY2024 revenue of ₹1,53,670 Cr) and shaped the design: year-over-year growth is available, multi-year trends are not.

4 Data Preparation with Python

A single reusable script, `01_clean_and_reshape.py`, reads the CSVs, converts the wide layout into a company-by-year layout, derives analysis fields and writes five SQL-ready files plus a data-quality report.

Output tables

Table	Grain	Rows	Purpose
dim_company	One row per company	4,668	Name, codes, industry, market cap, size bucket, financial-sector flag
fact_pnl	Company x fiscal year	9,330	Sales, profit, margins, interest coverage
fact_cashflow	Company x fiscal year	9,330	Operating, investing, financing, free cash flow
fact_balance_sheet	One row per company	4,668	Debt, net worth, liquidity and leverage ratios
fact_ratios	One row per company	4,668	ROE, ROCE, P/E, P/B, Altman Z and health category

Derived fields

- **Margins** recomputed from sales and profit, with NULL (not zero) when sales are zero or missing, which avoids divide-by-zero artefacts.
- **Interest coverage** = EBIT / interest; **net worth** = equity capital + reserves; calculated current ratio and debt-to-equity.
- **Financial health** from the Altman Z-score using standard cut-offs: below 1.81 Distress, 1.81 to 2.99 Grey Zone, above 2.99 Safe.
- **Size bucket** from market cap: Micro (<₹500 Cr), Small (₹500 to 5K Cr), Mid (₹5K to 20K Cr), Large (>₹20K Cr).
- **Financial-sector flag** for banks, NBFCs and insurers, whose leverage and liquidity ratios are not comparable.

Data-quality decisions

Issue found	Treatment
Wide layout with 'last year' / 'preceding year' columns	Reshaped to a long company x fiscal-year table
Only two annual periods available	YoY growth supported; 3-year trends explicitly out of scope
Three DVR share classes with no financial data	Removed from fact tables (6 rows), kept in the company dimension
Zero or missing sales	Margins set to NULL instead of zero or infinity
Extreme ROE / ROCE values (ROE ranges from -3,559% to 13,100%)	Industry comparisons use **medians**, not averages
Banks and NBFCs distort leverage and liquidity ratios	Flagged, and excluded from ratio-based pages by default

5 SQL Server Layer

The cleaned CSVs are loaded into a `FinancialAnalytics` database with typed columns, primary keys and foreign keys from every fact table to `dim_company`. Row counts were validated after load: 4,668 / 9,330 / 9,330 / 4,668 / 4,668.

Analysis queries

#	Business question	SQL concepts
B1	Executive KPIs: sales, profit, net margin	Aggregates, joins, filters
B2	Fastest sales growth (companies above ₹1,000 Cr)	LAG() window function
B3	Industry ranking by sales and by margin	CTE, RANK() OVER
B4	Top 10 companies by net profit	TOP, joins
B5	Top company inside each industry	ROW_NUMBER() OVER (PARTITION BY)
B6	High-debt, weak-coverage companies	Multi-table JOIN, compound filters
B7	Financial-health mix by size bucket	GROUP BY, conditional filters
B8	Profit reported but operating cash flow negative	Earnings-quality screen
B9	Industry ROE and ROCE (10+ companies)	GROUP BY, HAVING

Views for Power BI

- `vw_company_summary`: one row per company with latest-year P&L, YoY growth, balance sheet, cash flow and ratios (4,665 rows).
- `vw_financial_trend`: company x year P&L with cash flow (9,330 rows).
- `vw_industry_summary`: industry-level totals, margin, average ROE and loss-making count.

Example: year-over-year growth with LAG()

```
WITH yoy AS (
  SELECT company_id, fiscal_year, sales,
         LAG(sales) OVER (PARTITION BY company_id
                          ORDER BY fiscal_year) AS prev_sales
  FROM fact_pnl)
SELECT TOP 20 d.company_name, d.industry, y.sales, y.prev_sales,
       ROUND((y.sales - y.prev_sales) / y.prev_sales * 100, 2) AS sales_yoy_pct
FROM yoy y JOIN dim_company d ON d.company_id = y.company_id
WHERE y.fiscal_year = d.latest_fy
  AND y.prev_sales > 0 AND y.sales >= 1000
ORDER BY sales_yoy_pct DESC;
```

The model imported by Power BI uses the five SQL tables directly; the views support ad-hoc analysis and alternative reporting layers.

6 Data Model and DAX

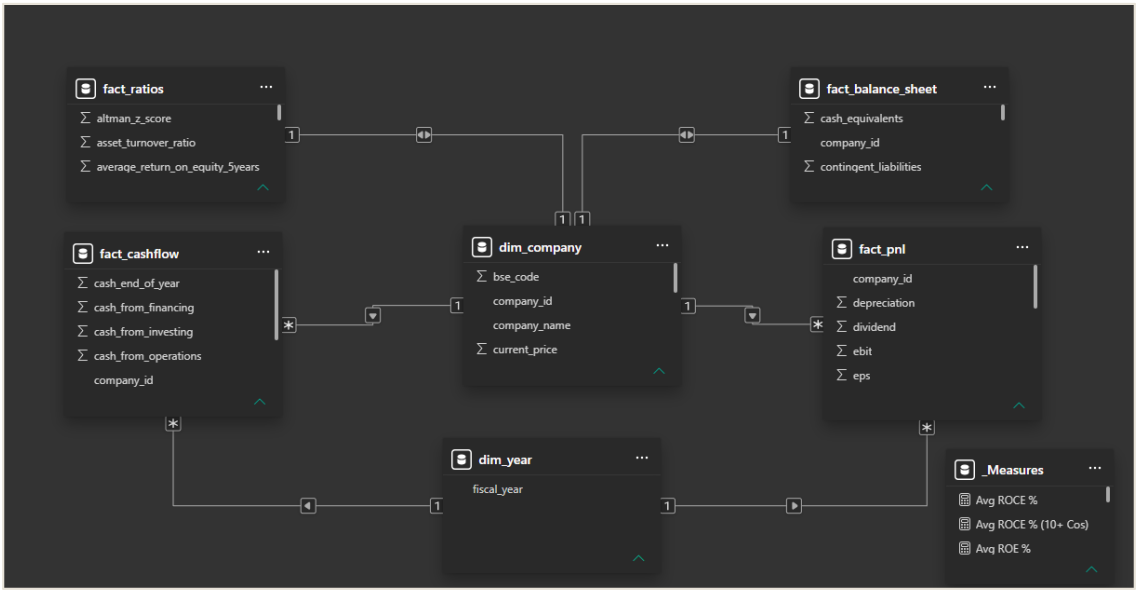


Figure 1: Power BI data model (star schema with a shared company dimension and a year dimension)

dim_company sits at the centre. **fact_pnl** and **fact_cashflow** are many-to-one to it and also relate to a small **dim_year** table for year slicing. **fact_balance_sheet** and **fact_ratios** are latest-year snapshots with one row per company, so they relate one-to-one and carry no year column.

DAX design

Group	Examples	Design note
Core P&L	Total Sales, Net Profit, Net Margin %, Operating Margin %	DIVIDE() for safe ratios
Time comparison	Sales PY, Sales YoY %, Profit YoY %	ALL(dim_year) with MAX(fiscal_year) - 1; Year slicer is single-select
Risk	Median Debt to Equity, Median Interest Coverage, Distress %	Medians resist extreme outliers
Cash flow	CFO, FCF, Cash Conversion %	Cash conversion = CFO / net profit
Valuation	Median PE, Median PB	Computed per industry and per size bucket
Robust Top-N	Net Margin % (5+ Cos), Median ROE % (10+ Cos)	Threshold built into the measure, see below
Display	Sales (L Cr), Net Profit (L Cr), Debt (L Cr)	Lakh-Crore scale with custom format strings

Two design problems worth noting

Top-N filters run before other filters. A chart showing the top 10 industries by margin, filtered afterwards to industries with at least 5 companies, ended up with only 5 bars. The fix was to put the condition inside the measure, so ineligible industries return blank and are never ranked:

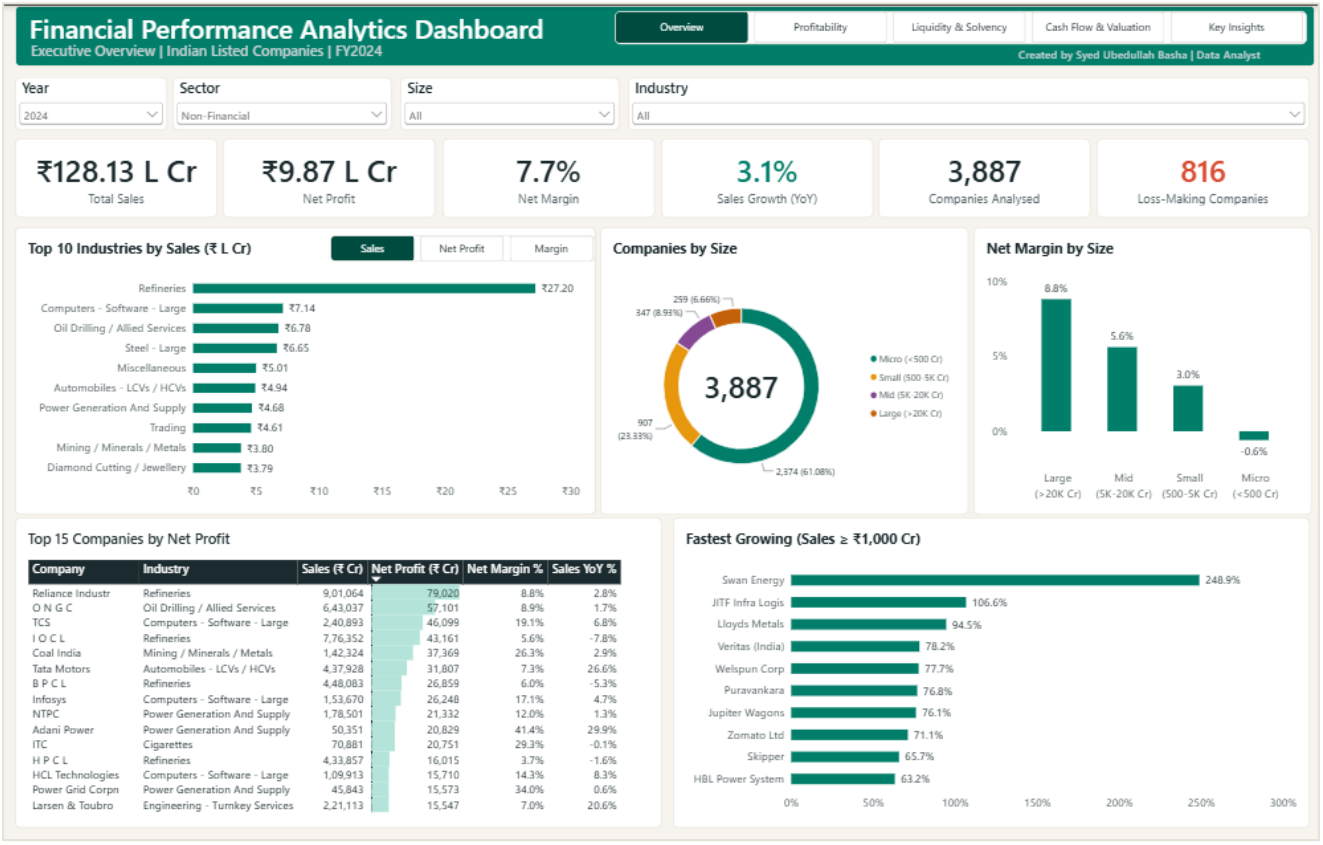
```
Net Margin % (5+ Cos) = IF ( [Companies] >= 5, [Net Margin %] )

Health Companies =
COUNTROWS ( FILTER ( fact_ratios,
    fact_ratios[financial_health] IN { "Safe", "Grey Zone", "Distress" } ) )
```

Unit consistency. ROE and ROCE are stored as percentage numbers (15.5 means 15.5%), while margins are fractions. Formatting each measure correctly prevented values such as 1,550% appearing on the dashboard.

7 Dashboard Walkthrough

The dashboard uses a custom **Emerald and Graphite** theme on a 1750 x 1100 canvas. A page navigator, synced slicers (Year, Sector, Size, Industry) and bookmark-based toggle buttons let the reader switch between charts on the same page. Defaults are **FY2024** and **Non-Financial** companies.



Page 1: Overview

What it shows

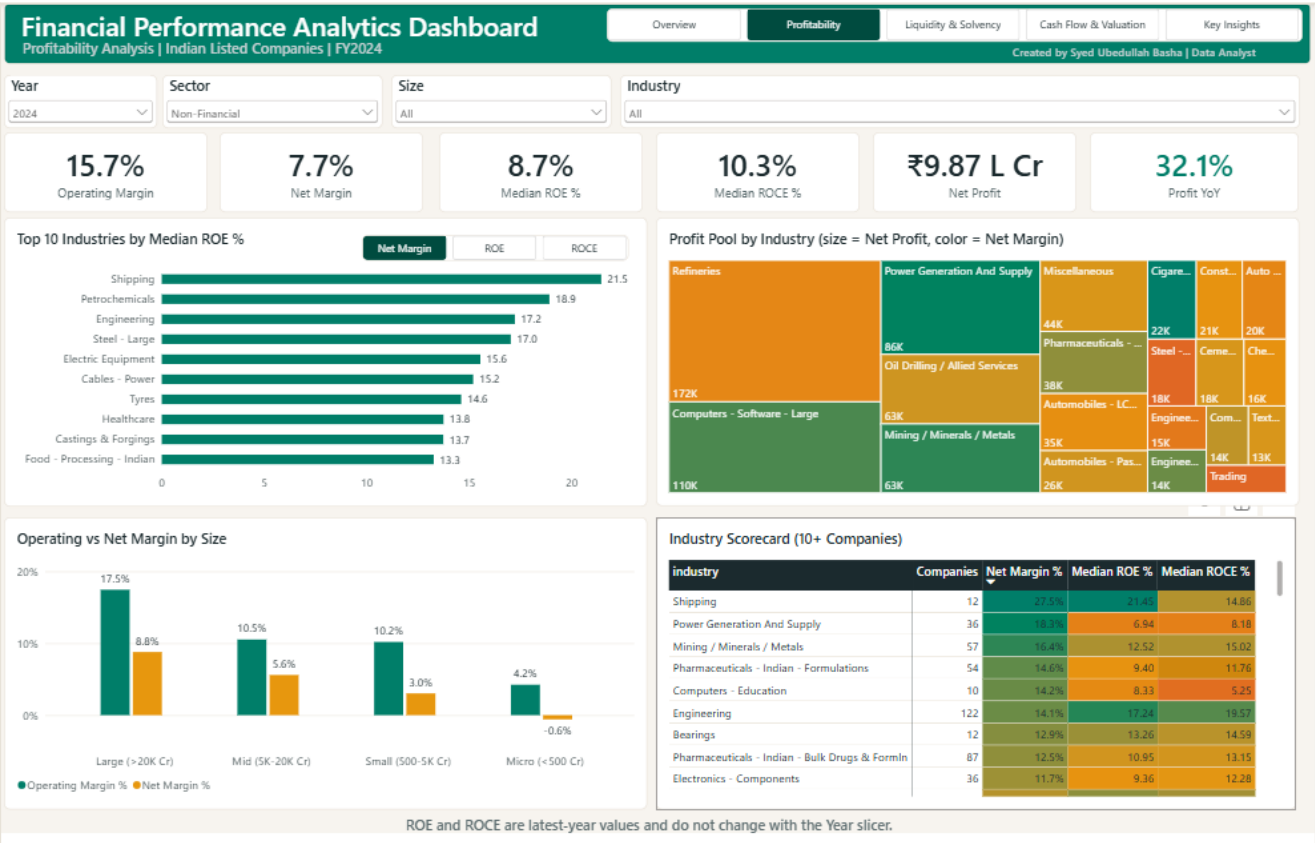
- Six KPI cards: sales, net profit, margin, growth, companies, loss-making count
- Top 10 industries with a Sales / Net Profit / Margin toggle
- Company mix and net margin by size bucket
- Top 15 companies by profit and fastest-growing companies (sales of ₹1,000 Cr or more)

Key insights

- ₹128.13 L Cr sales and ₹9.87 L Cr profit (7.7% margin); like-for-like sales growth is 3.1%
- Refineries is the largest industry: ₹27.2 L Cr sales and ₹1.72 L Cr profit
- Large caps earn a net margin of 8.8%; micro caps lose 0.6%
- 816 companies (21%) are loss-making

Analyst note: the Fastest Growing chart only ranks companies whose sales were at least ₹1,000 Cr in both years, which avoids extreme percentages from tiny bases. Year-over-year figures compare companies whose latest reported year is FY2024, so growth is like-for-like.

Interaction design: a page navigator on every page, slicers synced across pages, single-select Year (so YoY always compares two adjacent years), and bookmark buttons that switch between three charts in the same space on Pages 1 to 4.



Page 2: Profitability

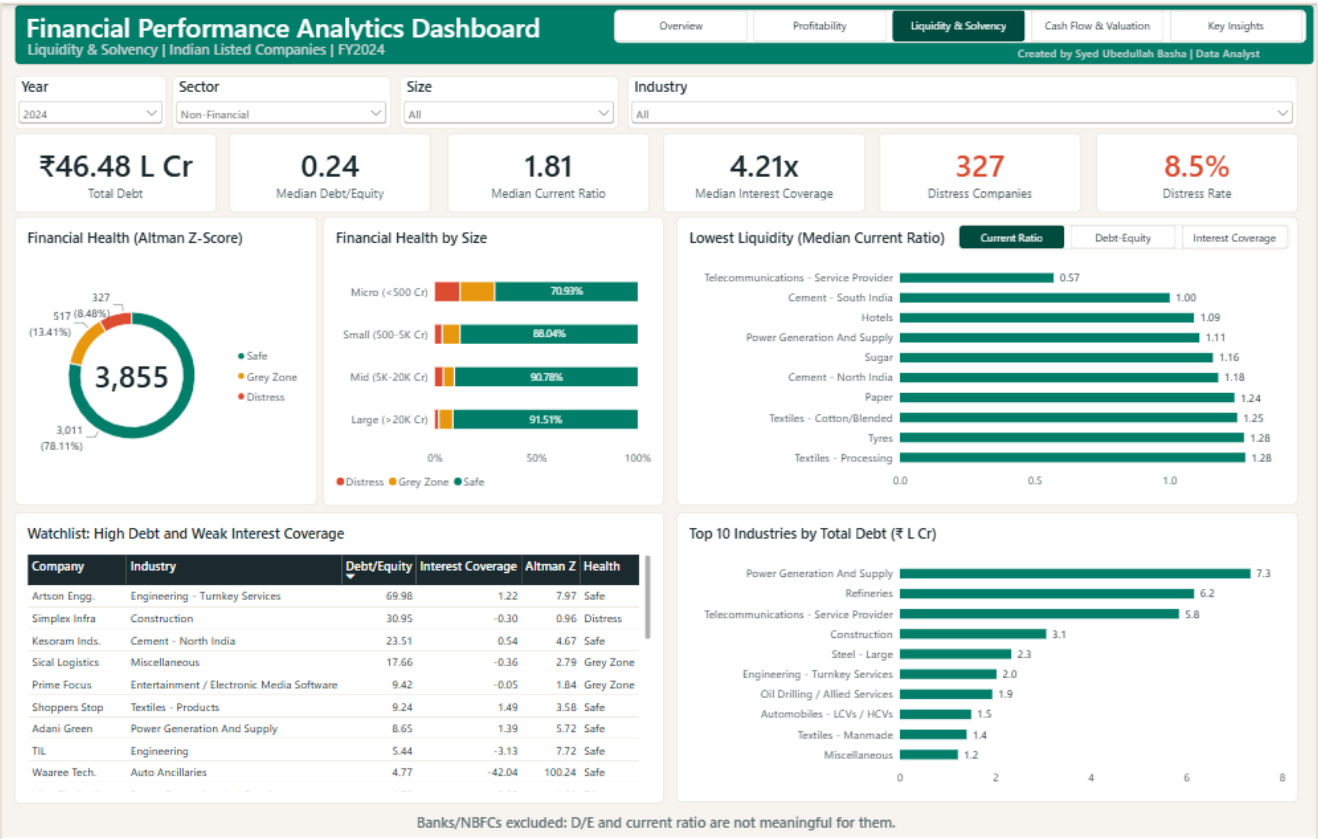
What it shows

- Operating and net margin, median ROE and ROCE, profit growth
- Top industries by net margin, ROE or ROCE (toggle)
- Profit-pool treemap: box size is net profit, colour is net margin
- Operating versus net margin by size, and an industry scorecard heat map

Key insights

- Operating margin is 15.7% but net margin only 7.7%: interest, depreciation and tax absorb about half of operating profit
- Among industries with 10+ companies, Shipping (27.5%), Power Generation (18.3%) and Mining (16.4%) earn the highest net margins
- Median ROE is 8.7% and median ROCE 10.3% (latest-year values)
- Refineries (₹1.72 L Cr), Software (₹1.10 L Cr) and Power Generation (₹0.86 L Cr) form the largest profit pools

Analyst note: ROE and ROCE are stored as percentage numbers and shown with a % format string; industry values are **medians** and only industries with 10 or more companies are ranked. The treemap and scorecard use a three-point colour scale (coral to amber to emerald) anchored at fixed values so a few outliers cannot wash out the colours.



Page 3: Liquidity and Solvency

What it shows

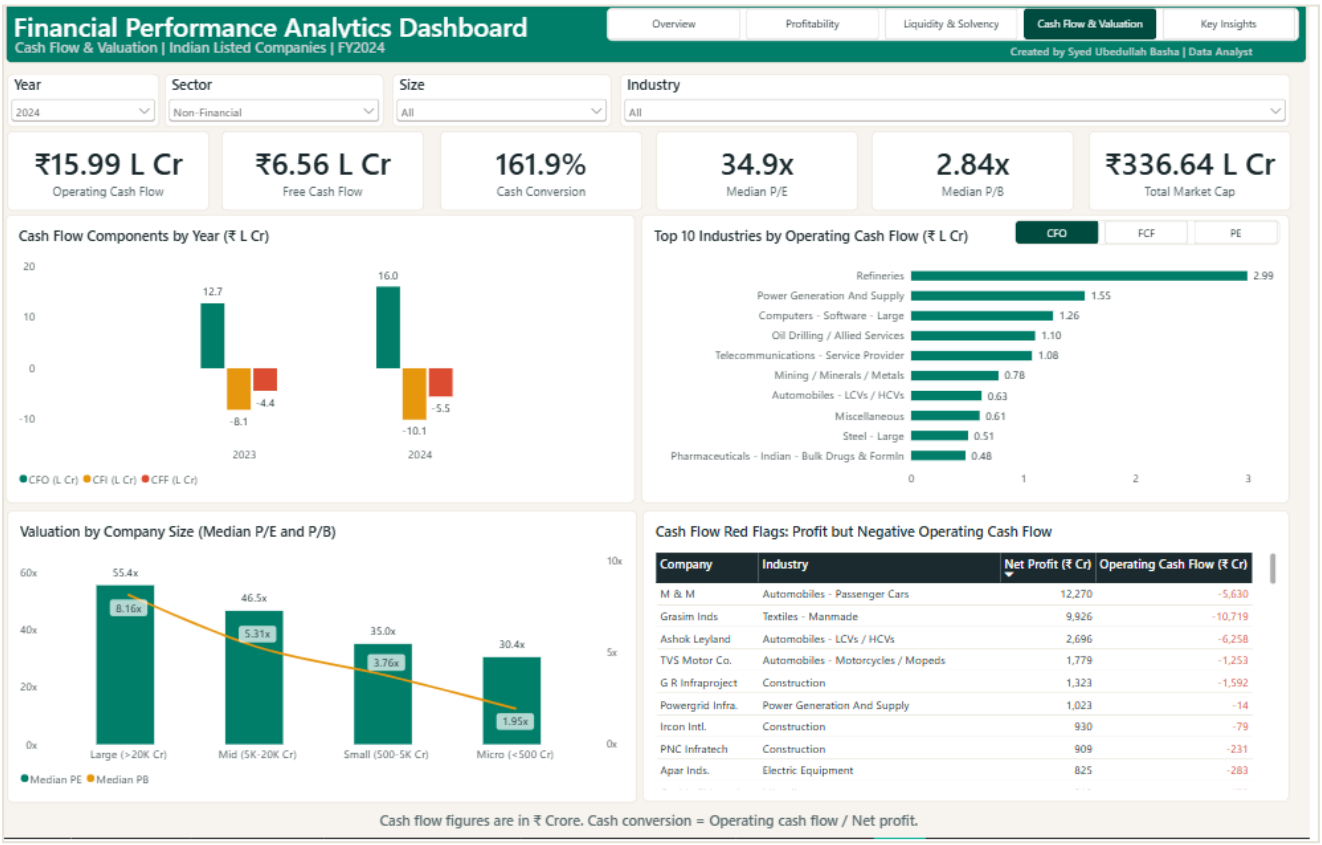
- Total debt, median debt-to-equity, current ratio, interest coverage, distress count and rate
- Altman Z health mix overall and by size bucket
- Weakest industries by liquidity, leverage and coverage (toggle)
- Watchlist of high-debt, weak-coverage companies and top industries by total debt

Key insights

- Median debt-to-equity 0.24 and current ratio 1.81 point to conservative financing
- 3,011 companies (78.1%) are Safe, 517 Grey Zone, 327 (8.5%) Distress
- Health improves with size: 91.5% of large caps are Safe versus 70.9% of micro caps
- Total debt is ₹46.48 L Cr, led by Power Generation (₹7.3 L Cr), Refineries (₹6.2 L Cr) and Telecom Services (₹5.8 L Cr)

Banks and NBFCs are excluded on this page because debt-to-equity and current ratio are not meaningful for them.

Analyst note: the watchlist flags non-financial companies with market cap of at least ₹500 Cr, debt-to-equity above 2 and interest coverage below 1.5. The Altman Z-score uses a different formula, so a flagged company can still show a Safe health category; the two views complement each other.



Page 4: Cash Flow and Valuation

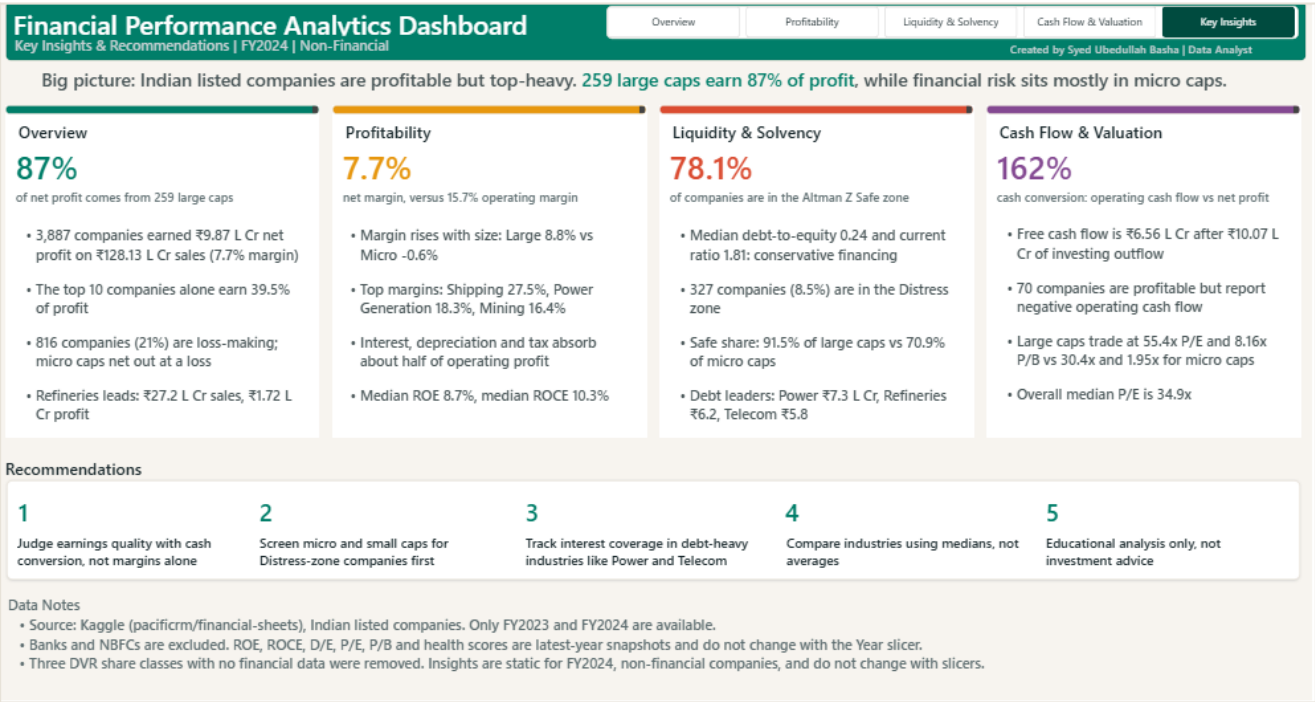
What it shows

- Operating and free cash flow, cash conversion, median P/E and P/B, total market cap
- Cash-flow components for FY2023 and FY2024
- Top industries by operating cash flow, free cash flow or P/E (toggle)
- Valuation by size, and a red-flag table of profit without operating cash

Key insights

- Operating cash flow of ₹15.99 L Cr is 162% of net profit; free cash flow is ₹6.56 L Cr
- Investing (-₹10.1 L Cr) and financing (-₹5.5 L Cr) outflows show heavy reinvestment alongside debt repayment and payouts
- 70 companies earn more than ₹100 Cr profit but report negative operating cash flow (for example M&M and Grasim)
- Large caps trade at 55.4x P/E and 8.16x P/B versus 30.4x and 1.95x for micro caps; overall median P/E is 34.9x

Analyst note: the cash-flow-by-year chart is disconnected from the Year slicer so FY2023 and FY2024 always appear side by side. A negative operating cash flow is not always a warning sign: companies with large lending or finance arms, such as M&M and Grasim, often report it, so the red-flag table is a screen for review, not a verdict.



Page 5: Key Insights and Recommendations

The final page summarises the story in one banner message, four insight cards (each with a headline number) and five recommendations, followed by data notes so that every limitation is visible to the reader.

8 Recommendations

1. **Judge earnings quality with cash conversion**, not margins alone. Profit that does not turn into operating cash deserves a closer look.
2. **Screen micro and small caps for Distress-zone companies first**, since financial risk is concentrated there.
3. **Track interest coverage in debt-heavy industries** such as Power Generation and Telecom Services.
4. **Compare industries using medians**, because averages are distorted by extreme outliers.
5. **Treat this as an educational analysis**, not investment advice.

9 Limitations and Assumptions

- **Two years only.** The dataset has FY2023 and FY2024 annual results, so multi-year trends are not covered.
- **Snapshot ratios.** ROE, ROCE, debt-to-equity, P/E, P/B and health scores are latest-year values and do not change with the Year slicer.
- **Financial sector excluded** from ratio analysis by default; the Sector slicer can include it, but ratios are then not comparable.
- **Like-for-like comparison.** Year-over-year views use companies whose latest reported year is FY2024 (3,887 non-financial companies).
- **Static insights.** The written insights on Page 5 are calculated for FY2024, non-financial companies, and do not change with slicers.
- **Source reliability.** Figures come from a public Kaggle dataset and were not independently verified against company filings.

10 Skills Demonstrated

Skill	Evidence in this project
Data cleaning (Python, pandas)	Wide-to-long reshape, derived metrics, outlier and missing-value decisions, quality report
SQL Server	Schema design, keys, BULK INSERT load, window functions, CTEs, views, row-count validation
Data modelling	Star schema with shared dimensions, snapshot versus time-series fact tables
DAX	Time comparison, medians, threshold-based Top-N measures, custom number formats
Dashboard design	Custom theme, consistent grid, navigator, synced slicers, bookmark toggles, conditional formatting
Analytical thinking	Questions defined first, caveats stated, findings tied to numbers and recommendations
Problem solving	Diagnosed Top-N ordering, unit mismatch, hidden two-year limit and bad join keys before they reached the dashboard

11 Repository Structure and Reproduction

```
financial-performance-analytics-dashboard/
├── README.md
├── LICENSE
├── .gitignore
├── data/
│   └── README.md
├── python/
│   └── 01_clean_and_reshape.py
├── sql/
│   ├── 02_create_tables_and_load.sql
│   └── 03_analysis_queries_and_views.sql
├── powerbi/
│   └── Financial_Performance_Analytics_Dashboard.pbix
├── report/
│   └── Financial_Performance_Analytics_Report.pdf
└── screenshots/
    └── 01_Overview.png ... 06_Data_Model.png
```

How to reproduce

1. Download the dataset from Kaggle (pacificrm/financial-sheets, see data/README.md) and unzip the 11 CSV files into data/raw/.
2. Set `IN_DIR` in `python/01_clean_and_reshape.py` and run it. It writes the five clean CSVs and a data-quality report.
3. Run `02_create_tables_and_load.sql` in SSMS (update the file paths in BULK INSERT, or use the Import Wizard) and confirm the row counts.
4. Run `03_analysis_queries_and_views.sql` to explore the analysis and create the views.
5. Open the .pbix file, point the data source to your SQL Server database, refresh, and set Year to 2024 and Sector to Non-Financial.

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